First stochastic SPH run: ideal symplectic step + per-particle Metropolis sweep

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Abstract

Working notes for the first stochastic SPH run. One deterministic *ideal* symplectic step is followed by a single Metropolis sweep over the *particles*. Each particle move draws one Gaussian stress ξ_a at particle a, applies it as momentum- and energy-conserving kicks through *all* of a's bonds, and is accepted with $\min(1, e^{\Delta \mathbb{S}_a})$, where $\Delta \mathbb{S}_a = \Delta t\, V_a\, \xi_a : \varepsilon_a$ sees the full SPH strain. *All* dissipation comes from the accept/reject asymmetry; there is no separate deterministic dissipative step. The mean drift of the sweep is *exactly* the deployed two-level viscous force `dissipative_force!`, and the noise covariance is forced by the match — there is no free amplitude and nothing to calibrate. Derivations, accuracy limits, and the implementation plan are collected in the appendices; the end target is the stochastic decay of a 2D sound wave.

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I. SCOPE AND NOTATION

Notation follows `sph2.tex`: particle a carries the extensive entropy s_a and momentum m_a^i; the smoothed entropy field is $S(x_a) = \sum_b s_b W(x_a, x_b)$; the EOS solve yields v_a, γ_a, $T_{\mathrm{rel},a}$; and the smoothed conjugates give the velocity $\dot{x}_a$, the temperature T_a, and

$$\mathsf{b}_a \equiv \frac{\dot{x}_a}{\mathsf{T}_a}\,. \tag{1}$$

$D_j(x_a, x_b) \equiv \partial W(x_a, x_b)/\partial x_a^j$ is the kernel gradient, with $D_j(x_b, x_a) = -D_j(x_a, x_b)$, and $V_a \equiv s_a/S_a$ is the particle "volume" (the SPH estimate of $\mathrm{d}^d x$, `sph2.tex` App. D).

Three results from `sph2.tex` are used verbatim :

- Density-frame noise covariance (FDT):

$$\left\langle \xi^{ij}(\boldsymbol{x},t)\,\xi^{lm}(\boldsymbol{x}',t')\right\rangle = 2\,T_{\rm rel}\,\kappa^{ijlm}(v)\;\delta(t-t')\,\delta^2(\boldsymbol{x}-\boldsymbol{x}')\,. \tag{2}$$

- Tthe entropy change that keeps $\mathbb{H}$ fixed under a momentum kick Δm_a (to linear order; App. C):

$$\Delta s_a = -\,\mathsf{b}_a\cdot\Delta m_a\,. \tag{3}$$

- Total entropy change:

$$\Delta\mathbb{S} = \sum_a \Delta s_a = -\sum_a \mathsf{b}_a\cdot\Delta m_a\,. \tag{4}$$

For deterministic hydro the viscous force was computed in two passes (`sph2.tex` App. D, `dissipative_force!`): Pass 1 builds the SPH strain

$$\varepsilon_{ji}(x_a) \;=\; \frac{1}{S(x_a)}\sum_b s_b\left(\mathsf{b}_{i,b}-\mathsf{b}_{i,a}\right)D_j(x_a,x_b) \tag{5}$$

and Pass 2 assembles the viscous force

$$f^i_{\rm det}(x_a) = \sum_b s_a s_b\,D_j(x_a,x_b)\left[\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_a\varepsilon_{lm}(x_a)+\left.\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\right|_b\varepsilon_{lm}(x_b)\right]. \tag{6}$$

We will show how the stochastic step reproduces this mean strain

II. THE STOCHASTIC STEP

A full step takes $(x_a^n, m_a^n, s_a^n) \to (x_a^{n+1}, m_a^{n+1}, s_a^{n+1})$ by an ideal sub-step followed by one per-particle sweep.

a. A — ideal symplectic sub-step. Advance (x_a, m_a) by one step of the *ideal* Hamiltonian flow with the 2nd-order symplectic integrator at fixed Δt (acoustic CFL). No dissipative or stochastic term here: s_a is unchanged and $\mathbb{S}$, $\mathbb{H}$ are conserved to integrator order.

b. B — freeze and strain. Freeze a snapshot of the coefficients $\{S_a, s_a, \mathsf{b}_a, v_a, \gamma_a, T_{{\rm rel},a}, \kappa|_a\}$ and precompute the strain ε_a (5) at every particle. During the sweep only the accumulated increments $(\Delta m_a, \Delta s_a)$ change; they are applied to (m_a, s_a) once, at the end of the sweep, and the coefficients are then refreshed for the next step.

c. C — the proposal at particle a. For particle a, draw one symmetric 2×2 Gaussian stress $\xi_a^{ij} = \xi_a^{ji}$, mean zero, with covariance

$$\boxed{\left\langle \xi_a^{ij}\,\xi_a^{lm}\right\rangle \;=\; \frac{2\,T_{{\rm rel},a}\,\kappa^{ijlm}(v_a)}{\Delta t\,V_a}\,,\qquad V_a\equiv\frac{s_a}{S_a}} \tag{7}$$

This is the discretised FDT (2) with $\delta(t)\to 1/\Delta t$ and $\delta^d(x)\to 1/V_a$. From ξ_a build a *proposal*: a trial update of a and all of its neighbours, patterned on the particle-a stress

piece of the viscous force (6). The proposed momentum changes are, for every bond (a, b) with $r_{ab} < 2h$,

$$\boxed{\Delta m_a \mathrel{+}= \Delta t\, \frac{s_a s_b}{S_a^2}\, \xi_a^{ij} D_j(x_a, x_b)\,, \qquad \Delta m_b \mathrel{-}= \Delta t\, \frac{s_a s_b}{S_a^2}\, \xi_a^{ij} D_j(x_a, x_b)\,,} \tag{8}$$

together with $\Delta s_c = -\mathsf{b}_c \cdot \Delta m_c$ (3), for every particle c neighboring a. The proposal conserves momentum bond by bond, conserves $\mathbb{H}$ to linear order in the kick, and has $\Delta m \sim \sqrt{\Delta t}$; positions are untouched. Nothing is applied yet — whether the trial update is kept is decided next.

d. D — accept / reject. The entropy change of the proposed update collapses onto the strain:

$$\begin{aligned} \Delta \mathbb{S}_a &= \sum_c \Delta s_c \qquad (9)\\ &= -\sum_c \mathsf{b}_c \cdot \Delta m_c = \Delta t\, \frac{s_a}{S_a^2}\, \xi_a^{ij} \sum_b s_b \big(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}\big) D_j(x_a, x_b) \\ &= \boxed{\Delta t\; V_a\; \xi_a^{ij}\, \varepsilon_{ji}(x_a)} \qquad (10) \end{aligned}$$

This is the discrete form of $\int \mathrm{d}t\, \mathrm{d}^d x\; \xi^{ij} \partial_j \beta_i$, and of either sign. Note to compute $\Delta \mathbb{S}_a$ at this linearized level we can just the previously stored strain. Accept the proposal with probability

$$\boxed{P_{\mathrm{acc}}(a) = \min\big(1,\; e^{\,\Delta \mathbb{S}_a}\big)\,;} \tag{11}$$

on acceptance the trial $(\Delta m, \Delta s)$ are added to the accumulators; on rejection the proposal is discarded entirely and the state is unchanged. Because $\xi_a \to -\xi_a$ is equiprobable and the covariance depends only on frozen coefficients, this is a plain symmetric-proposal Metropolis move: it preserves the target weight $\propto e^{\mathbb{S}}$ on the $(\mathbb{H}, \mathbf{P})$ level set fixed by the ideal flow. Dissipation is entirely the acceptance asymmetry.

e. E — the sweep. Zero the increment accumulators $(\Delta m_a, \Delta s_a)$, then one pass over all particles $a = 1 \ldots N$, each visited once, accumulating accepted kicks; then fold the accumulators into (m_a, s_a) and refresh. With the freeze the moves are mutually independent, so the sweep is order-free and parallel (App. B). Exactly one sweep per ideal step: the covariance (7) carries the Δt.

III. MEAN DRIFT

Δm and $\Delta \mathbb{S}_a$ are both linear in the Gaussian ξ_a, so the Metropolis average follows from the identity $\big\langle X \min(1, e^Y) \big\rangle = \frac{1}{2} \langle XY \rangle + O(\sigma^3)$ for mean-zero jointly Gaussian (X, Y). Applied to a single particle move, and summing the sweep — particle a is kicked once by its *own* move (the particle-a stress term) and once, in reaction, by *each neighbour's* move (their particle-b stress terms; the draws are independent, so the means add) — the result is (App. A)

$$\boxed{\big\langle \Delta m_a \big\rangle_{\mathrm{acc}} = \Delta t \sum_b s_a s_b\, D_j(x_a, x_b) \left[\frac{T_{\mathrm{rel}} \kappa^{ijlm}}{S^2} \bigg|_a \varepsilon_{lm}(x_a) + \frac{T_{\mathrm{rel}} \kappa^{ijlm}}{S^2} \bigg|_b \varepsilon_{lm}(x_b) \right] = \Delta t\, f^i_{\mathrm{det}}(x_a)\,,} \tag{12}$$

the **full** two-level force (6). Two cancellations in the derivation are structural: the Metropolis $\frac{1}{2}$ (only the downhill half of the noise is suppressed) exactly eats the FDT factor of 2 in (7), and the $1/V_a$ from the discretised $\delta^d(x)$ supplies the volume power that a naive per-bond force misses (App. D). The covariance (7) is therefore *forced* by the match: there is no free amplitude, and the shear-wave and sound-wave runs are *validations*, not calibrations.

IV. DIAGNOSTICS — WHAT "WORKING" MEANS

- **Momentum:** $\sum_a m_a^i$ constant to machine precision.
- **Energy:** $\mathbb{H}$ shows the *expected* small positive drift at relative rate $\sim 1/(s_a \tau_{\rm visc})$ [App. C] on top of the integrator error — linear in t, with the predicted magnitude; anything faster is a bug.
- **Second law:** $\langle \Delta \mathbb{S} \rangle \geq 0$ per sweep; individual moves go both ways.
- **Acceptance:** near unity ($\gtrsim 0.99$) in the small-kick regime; it is *not* tuned. Low acceptance means Δt is too large for the given η/s and gradients.
- **Grid-scale modes:** rms(m_a) on a quiescent uniform background must *saturate* at the FDT equipartition value ($\langle v_a^2 \rangle \sim 1/s_a$; requires $s_a \gtrsim 10^3$ for non-relativistic thermal fluctuations), in contrast to the per-bond $\sqrt{t}$ runaway (App. D).
- **Equilibrium fluctuations:** time-averaged $\langle \delta S^2 \rangle$, $\langle m^2 \rangle$ vs. the FD prediction; repeat at $\Delta t/2$ to bound the $O(\Delta t)$ weak bias [App. C].
- **Mean drift:** sweep-averaged $\Delta m_a/\Delta t$ on a frozen sheared state matches `dissipative_force!` — (12) head-to-head.
- **Shear wave (validation):** decay rate of $M^y = A \sin(kx)$ matches $\gamma_k = (\eta/s)\, k^2/T_0$ with no tuned parameter.
- $\eta/s \to 0$**:** $\xi_a \to 0$; the run reduces to the deterministic ideal symplectic integrator.

Appendix A: Derivation of the mean drift

Write the kick to a from its own move as $X = \Delta m_a^i$ and the accept exponent as $Y = \Delta \mathbb{S}_a$; both are linear in ξ_a with frozen coefficients, so (X, Y) are jointly Gaussian and mean zero, and

$$\left\langle X \, \min(1, e^Y) \right\rangle = \tfrac{1}{2} \langle XY \rangle + O(\sigma^3) \tag{A1}$$

[split $\min(1, e^Y) = 1 + (e^Y - 1)\mathbb{1}[Y < 0]$ and use $\langle XY\, \mathbb{1}[Y<0] \rangle = \frac{1}{2}\langle XY \rangle$ for a mean-zero Gaussian pair; the remainder is $O(\sigma_Y^3) = O(\Delta t^{3/2})$]. With $G_j \equiv \sum_b s_b D_j(x_a, x_b)$ this gives

$$\left\langle \Delta m_a^i \right\rangle_{\rm acc} = \tfrac{1}{2}\, \Delta t^2\, \frac{s_a^2}{S_a^3}\, G_j\, \varepsilon_{lk}(x_a)\, \left\langle \xi_a^{ij} \xi_a^{kl} \right\rangle + O(\xi^3)\,. \tag{A2}$$

Substitute the covariance (7) with index pairs (ij), (kl),

$$\left\langle \xi_a^{ij} \xi_a^{kl} \right\rangle = \frac{2\, T_{\mathrm{rel},a}\, \kappa^{ijkl}\big|_a}{\Delta t\, V_a}\,, \qquad V_a = \frac{s_a}{S_a}\,, \tag{A3}$$

and collect the prefactors:

$$\begin{aligned} \left\langle \Delta m_a^i \right\rangle_{\mathrm{acc}} &= \underbrace{\tfrac{1}{2}\cdot 2}_{=1}\, \underbrace{\frac{\Delta t^2}{\Delta t}}_{=\Delta t}\, \underbrace{\frac{s_a^2}{S_a^3}\cdot\frac{S_a}{s_a}}_{=s_a/S_a^2}\, T_{\mathrm{rel},a}\, \kappa^{ijkl}\big|_a\, \varepsilon_{lk}(x_a)\, G_j \\ &= \Delta t\, \frac{s_a}{S_a^2}\, T_{\mathrm{rel},a}\, \kappa^{ijkl}\big|_a\, \varepsilon_{lk}(x_a)\, G_j\,. \end{aligned} \tag{A4}$$

The Metropolis $\frac{1}{2}$ cancels the FDT 2, and the $1/V_a$ removes one power of the particle volume.

Next the index contraction. In (A4) the strain carries the Pass-1 convention ε_{lk} = (gradient index first, velocity index second), so κ's *last* index l meets the gradient slot and its *third* index k the velocity slot — the opposite order from the deterministic contraction $\kappa^{ijlm}\varepsilon_{lm}$. One dummy relabel ($k \to m$) and the symmetry of κ in its last pair ($\kappa^{ijml} = \kappa^{ijlm}$, true of the $2\eta(\cdots) + \zeta(\cdots)$ structure) fix this:

$$\kappa^{ijkl}\, \varepsilon_{lk} \overset{k\to m}{=} \kappa^{ijml}\, \varepsilon_{lm} \;=\; \kappa^{ijlm}\, \varepsilon_{lm}\,; \tag{A5}$$

in particular only the *symmetric* part of ε survives, as in the deterministic force. With G_j restored,

$$\left\langle \Delta m_a^i \right\rangle_{\mathrm{acc}} = \Delta t\, \frac{s_a}{S_a^2}\, \left(T_{\mathrm{rel}} \kappa^{ijlm}\big|_a\, \varepsilon_{lm}(x_a)\right) \sum_b s_b\, D_j(x_a, x_b)\,, \tag{A6}$$

precisely the particle-a stress contribution to the two-level force (6).

This is half of the story: (A6) is the mean kick a receives *from its own move*. But a is also a neighbour of each b, so it is kicked again — in reaction — every time b does *its* move. Bond (b, a) is one of b's bonds, so b's move applies

$$\Delta m_a \mathrel{-}= \Delta t\, \frac{s_a s_b}{S_b^2}\, \xi_b^{ij}\, D_j(x_b, x_a) \;=\; +\Delta t\, \frac{s_a s_b}{S_b^2}\, \xi_b^{ij}\, D_j(x_a, x_b) \tag{A7}$$

(kernel antisymmetry), and this kick is accepted with b*'s* exponent $\Delta\mathcal{S}_b = \Delta t\, V_b\, \xi_b^{kl} \varepsilon_{lk}(x_b)$ — the same ξ_b in both, so (A2)–(A6) go through with every coefficient evaluated at b:

$$\left\langle \Delta m_a^i \right\rangle_{\mathrm{acc}}^{(b\text{'s move})} = \Delta t\; s_a s_b\, D_j(x_a, x_b)\, \frac{T_{\mathrm{rel}} \kappa^{ijlm}\big|_b\, \varepsilon_{lm}(x_b)}{S_b^2}\,. \tag{A8}$$

The draws ξ_a, $\{\xi_b\}$ are independent, so over one full sweep the means add — a is touched once as owner and once per neighbour as reactor — giving (12).

Appendix B: Sweep order, parallelism, and exactness

With the coefficient freeze and the accumulator fields — no in-place update of m, s — each particle's proposal ξ_a, its kicks, and its accept test depend only on frozen coefficients, the precomputed ε_a, and that particle's own draw (ξ_a, U_a) from a **particle-local RNG keyed by** $(\texttt{seed}, \texttt{step}, a)$ (not a shared stream; `seed` from `SimParams`, so the run is reproducible from its parameters). Nothing one move reads is written by another; the sole inter-move coupling is the additive accumulation into $(\Delta m, \Delta s)$, which commutes.

a. Order-independence. A single pass yields the same $(\Delta m, \Delta s)$ regardless of visit order — fixed, random-permuted, or fully parallel. The accepted set and every kick are *bit-identical* across orderings; the per-particle sums differ only by floating-point associativity ($\sim$ a few ulps).

b. One move per particle. Every particle is visited exactly once per sweep, so the noise/dissipation dose per Δt is spatially uniform by construction. Non-uniform density needs no attempt-weight correction: the covariance (7) carries $1/V_a$ and the strain (5) carries $1/S_a$, so the density dependence is exactly that of the deterministic force.

c. Synchronous (parallel) sweep. Decoupled decisions + additive writes: propose and decide all particles in parallel; reduce the accepted kicks (atomics or per-thread partials); apply. The keyed RNG makes every kick and accept flag independent of thread/rank count; only the reduction order (the last few ulps) changes. Distributed memory: standard MD halo — exchange positions and frozen coefficients within $2h$ of each subdomain boundary (the strain pass needs b within $2h$; the kicks reach $2h$), compute owned particles, reduce boundary contributions to owners.

d. Exactness: Metropolis-within-Gibbs. The frozen synchronous sweep is *not* an exact sampler of $e^{\mathbb{S}}$: each particle's accept test ignores the kicks applied by the other moves in the same sweep (their stencils overlap within $2h$), so the composite kernel preserves $e^{\mathbb{S}}$ only up to the standard $O(\Delta t)$ weak (Euler–Maruyama-type) error. The classical exact alternative is *sequential Metropolis-within-Gibbs*: visit particles in order, fold each accepted move into the live state and refresh b_c of the touched particles before the next move — each single-particle kernel then satisfies detailed balance and the composite preserves the target. But even this is exact only if the state dependence of the proposal covariance (7) ($T_{\mathrm{rel},a}$, $\kappa(v_a)$, V_a all move with the state) is included through the Gaussian Hastings ratio; dropping it is again an $O(\Delta t)$ bias (the usual multiplicative-noise ambiguity). Since *both* variants carry $O(\Delta t)$ weak bias, the frozen sweep — fully parallel and order-independent — loses nothing in accuracy order; the sequential + Hastings variant is the recorded upgrade path if precision equilibrium statistics are ever needed.

Appendix C: Known biases and their sizes

None of the following affects the sound-wave decay target at leading order; they are recorded so the diagnostics test for the *expected* deviations.

a. (a) Stationary distribution: $O(\Delta t)$ weak bias. From the coefficient freeze (App. B). Affects equilibrium fluctuation amplitudes at $O(\Delta t)$, not the mean drift. Detect by running the equilibrium test at Δt and $\Delta t/2$; cure, if ever needed, by sequential Metropolis-within-Gibbs with the Hastings ratio.

b. (b) Energy bookkeeping: the linearized Δs heats at $O(1/s_a)$. eq. (3) is the *first-order* term of "choose Δs so that $\mathbb{H}$ is conserved":

$$\Delta\mathbb{H} = \underbrace{\dot{x}_c \cdot \Delta m_c + \mathsf{T}_c\, \Delta s_c}_{=0} + \tfrac{1}{2}\, \Delta m \cdot \frac{\partial^2 \mathbb{H}}{\partial m^2} \cdot \Delta m + \dots \tag{C1}$$

Deterministically $\Delta m \sim \Delta t$ and the dropped term is a harmless $O(\Delta t^2)$ — which is why `dissipative_step!` gets away with the same formula. Stochastically $\Delta m \sim \sqrt{\Delta t}$, so

the dropped term is $O(\Delta t)$ per step and *sign-definite* (positive kinetic Hessian): every landed kick heats by $\sim \Delta m^2/2m_{\rm eff}$, which summed is the FDT noise-injection power. The friction work is booked into s correctly; the noise heating is booked nowhere, so $\mathbb{H}$ drifts monotonically upward at the relative rate

$$\frac{\dot{\mathbb{H}}}{\mathbb{H}} \sim \frac{1}{s_a\,\tau_{\rm visc}}\,, \tag{C2}$$

suppressed by the per-particle entropy $s_a \gtrsim 10^3$. Treat $\Delta\mathbb{H}(t)$ as an *expected-drift diagnostic*: small, positive, linear in t, with this rate — its magnitude is itself a check of the noise normalization. Upgrade if ever needed: midpoint evaluation $\Delta s_c = -\frac{1}{2}[\mathsf{b}_c(m) + \mathsf{b}_c(m + \Delta m)] \cdot \Delta m_c$ (residual $O(\Delta m^3)$), at the cost of a velocity re-solve per accepted move. The mean drift (12) is unaffected: a quadratic addition to $\Delta\mathbb{S}_a$ shifts the Metropolis mean only through vanishing Gaussian third moments.

c. (c) The realized increments are non-Gaussian — by design. The accepted kick distribution is the Gaussian proposal *tilted* by the acceptance: in the scalar toy ($\Delta\mathbb{S} = c\xi$, $\xi \sim N(0,\sigma^2)$) it is the untouched Gaussian on the favorable side, a Gaussian *shifted* to $c\sigma^2$ on the unfavorable side ($N(\xi)e^{c\xi} \propto e^{-(\xi - c\sigma^2)^2/2\sigma^2}$), plus a rejection point mass at zero. Its mean $\frac{1}{2}c\sigma^2$ *is* the drift — the tilt cannot be removed without removing the dissipation. Higher cumulants are $O(\Delta t^{3/2})$ and vanish in the diffusion limit. Additionally, the linearized $\Delta\mathbb{S}_a$ mis-weights rare large kicks by $e^{O(k^2\Delta t/\tau)}$ at k standard deviations, distorting only far tails ($k \sim \sqrt{\tau/\Delta t}$) that are never sampled. Only mean and covariance survive at $O(\Delta t)$ — both are exactly right by Sect. III.

Appendix D: Why not per-bond accept/reject (record)

The first design used one Metropolis test per *bond*: a pair kick $\delta_{ab} \propto \Delta t\,\mathcal{A}\sqrt{s_a s_b}\,(\xi_a/S_a + \xi_b/S_b)D(a,b)$ accepted with $\min(1, e^{-(\mathsf{b}_a - \mathsf{b}_b)\cdot\delta_{ab}})$. It was coded, tested, and deleted — it fails structurally; do not revisit:

- **The mean is the single-level force, always.** The accept probability sees only δ_{ab}, so $\langle\delta_{ab}\rangle_{\rm acc} \propto \langle\delta_{ab}\delta_{ab}\rangle\,\Delta\mathsf{b}^{(ab)}$ is a single-bond object with *coincident* kernel gradients $D_j(a,b)D_l(a,b)$ — never the two-level strain. No prefactor choice changes this.
- **The single-level force is dimensionally** 1/vol **too big:** $D_l(a,b)\,\Delta\mathsf{b}^{(ab)}$ is not a gradient estimator ($\sim [T^2]$ where $\partial_l\mathsf{b}$ is dimensionless); the amplitude $\mathcal{A}$ had to absorb a volume factor and be calibrated.
- **Odd-even runaway.** On a quiescent uniform background the per-bond kicks inject independent grid-scale momentum that the *smoothed* b_a cannot see, so the accept/reject provides no friction on those modes: $\mathrm{rms}(m_a) \propto \sqrt{t}$ without bound at every amplitude (observed), while the resolved (smoothed) sector stayed flat.

The per-particle move fixes all three at once: the accept test sees the full strain ε_a [(10)], the mean is the two-level force with the volume power correct (App. A), and the injected pattern $\xi_a\,D(x_a,x_b)$ is smooth in b (a kernel-gradient dipole), so the grid-scale modes are excited far less and the strain-based friction acts on what is excited.

Appendix E: Implementation

Reused unchanged: `sph.jl` (`Particle`, EOS, `kappa_tensor`), `celllist.jl` (`leapfrog_step!`, cell-list machinery, `dissipative_force!` Pass 1, `wendland_c2_grad`, `minimum_image`), `setup.jl` (IC helpers, `thermalize!`), `io.jl` (`SnapshotWriter`). New code lives in `stochastic.jl` + driver scripts.

a. Noise draw in code. `generate_noise` (`sph2.tex` App. E1) returns the unit-normalized stress $\hat{\xi}$ with $\left\langle \hat{\xi}^{ij}\hat{\xi}^{lm}\right\rangle = \kappa^{ijlm}(v)$ (*no* $T_{\rm rel}$); `draw_xi` rescales by $\sqrt{2T_{\rm rel}}$ to the continuum FDT normalization 2`kappa_tensor` (verified by MC audit, `teststochastic.jl` Step 2, at all v). The per-particle draw of (7) is then `draw_xi`$/\sqrt{\Delta t\, V_a}$.

b. The full loop.

1. IC: uniform background, small periodic box (`setup_ic` / `setup_shear_wave`); `thermalize!`.
2. Per step: `leapfrog_step!` → `rebuild_cell_list!` → refresh+freeze (`smooth_to_particles!`, `update_thermodynamics!`, `update_velocities!`) → strain pass → `particle_sweep!` → `apply_sweep!` → diagnostics.

c. Status. Done: accumulator fields + `apply_sweep!` (Step 0); the cell-list neighbour stencil — the traversal used by the strain pass and the sweep — validated against $O(N^2)$ brute force (Step 1; requires `ncx`, `ncy` ≥ 3); `draw_xi` + MC covariance audit (Step 2). The per-bond sweep and its bond list were coded, tested, and deleted (App. D).

```
particle_sweep!(particles, params, cl, eos, dt, key) -> (n_acc, n_try, ΔS_tot)
  # coefficients frozen; ε_a from the strain pass (dissipative_force! Pass 1)
  for a in eachindex(particles); particles[a].Δm = 0; particles[a].Δs = 0; end
  for a in eachindex(particles)
      rng = Xoshiro(hash((params.seed, key, a)))           # particle-local RNG
      Va  = particles[a].s / particles[a].S_smooth
      ξa  = draw_xi(rng, particles[a], eos, cs2v) / sqrt(dt * Va)   # cov: eq (persite_cov)
      ΔS  = dt * Va * sum(ξa[i,j]*ε_a[j,i] for i,j)         # = dt V_a ξ:ε   eq (dSa)
      n_try += 1
      if rand(rng) < exp(ΔS)                                # test BEFORE bond loop
          for b in neighbors(a), r_ab < 2h                  # cell-list stencil
              grad = wendland_c2_grad(x_a - x_b, ...)
              f    = dt * (s_a*s_b / S_a^2) * (ξa * grad)  # SVector{2}     eq (kicks)
              particles[a].Δm += f ; particles[a].Δs -= dot(b_vec[a], f)
              particles[b].Δm -= f ; particles[b].Δs += dot(b_vec[b], f)
          end
          n_acc += 1 ; ΔS_tot += ΔS
      end
  end
```

The accumulators are zeroed at entry (a full pass first — a particle is written as a neighbour of others' moves), so no caller-side reset is needed; `apply_sweep!` folds $(\Delta m, \Delta s)$ into (m, s) afterwards. The accept test needs only ξ_a and ε_a [(10)] — no neighbour loop — so rejected proposals cost $O(1)$. *Checks* (extend `teststochastic.jl`): (i) $\eta/s = 0 \Rightarrow$ all $\Delta m, \Delta s = 0$;

(ii) $|\sum_a \Delta m_a| / \sum_a |\Delta m_a| < 10^{-12}$; (iii) GENERIC degeneracy $\sum_a (\mathsf{T}_a \Delta s_a + \dot{x}_a \cdot \Delta m_a) = 0$ to round-off; (iv) particle-local RNG $\Rightarrow$ visit-order independence to $< 10^{-12}$; (v) the identity (10): the cheap $\Delta t V_a \, \xi : \varepsilon$ equals the explicit $-\sum_c \mathsf{b}_c \cdot \Delta m_c$ over the distributed kicks, per particle, to round-off; (vi) **mean drift:** many sweeps on one frozen sheared state, $\langle \Delta m_a \rangle / \Delta t$ vs. `dissipative_force!` — slope 1, the decisive test of (12).

e. Then. *Uniform-background stability* (`teststochastic_run.jl` rewired): $s_a \gtrsim 10^3$; $\mathrm{rms}(m_a)$ saturates at the FDT value; $\mathbb{H}$ drift positive at the App. C(b) rate; acceptance $\gtrsim 0.99$; repeat at $\Delta t/2$. *Shear-wave validation* (`teststochastic_shear.jl`): ensemble-averaged decay of $M^y = A \sin(kx)$ vs. $(\eta/s) k^2 / T_0$, no tuned parameter; an $\eta/s = 0$ run measures the numerical baseline. *2D sound-wave decay — the target* (`teststochastic_sound.jl`, built on `testsound_viscous.jl`): eigenmode IC with genuinely 2D $\vec{k}$ ($n_x = n_y = 1$); ensemble-mean amplitude decay vs. $e^{-\Gamma_s k^2 t/2}$, $\Gamma_s = (\frac{4}{3}\eta/s + \zeta/s)/T_{\mathrm{rel},0}$; final profile vs. `analytic_M`; no fitting. *Parallel sweep* (optional): parallel particle loop, reduction into the accumulators; bit-identical to serial for the same key (App. B).